

Monday, April 7

Probability Density, Survival, and Hazard Functions

Let T be a continuous random variable that is time-till-event. Four related functions are used to describe the distribution of T .

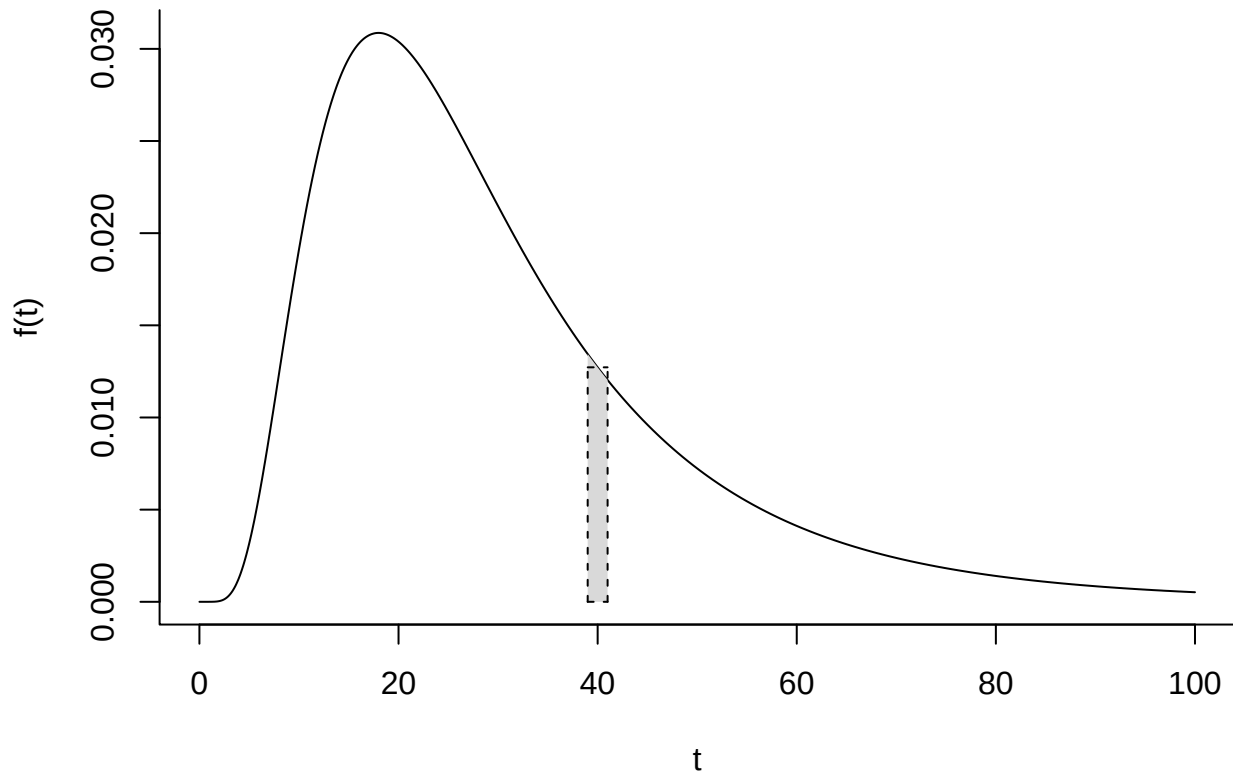
The Probability Density Function

The *probability density function* of T is

$$f(t) = \lim_{\delta t \rightarrow 0} \frac{P(t \leq T < t + \delta t)}{\delta t}.$$

If δ is relatively small then $P(t \leq T < t + \delta t) \approx f(t)\delta t$ and so $f(t) \approx P(t \leq T < t + \delta t)/(\delta t)$ and thus $f(t)$ is approximately *proportional* to the probability that T is between t and $t + \delta t$. So $f(t)$ is approximately *proportional to the probability that the event will happen “near” t* .

For the distribution below, the probability that T is *approximately* 40 (say, between 39 and 41) equals the area under the curve and between 39 and 41. This probability is *approximated* by the rectangle, which has area $wf(40)$, where $w = 2$ is the width of the rectangle and $f(40)$ is the height of the rectangle. So the probability that T is approximately 40 is *proportional to* $f(40)$.

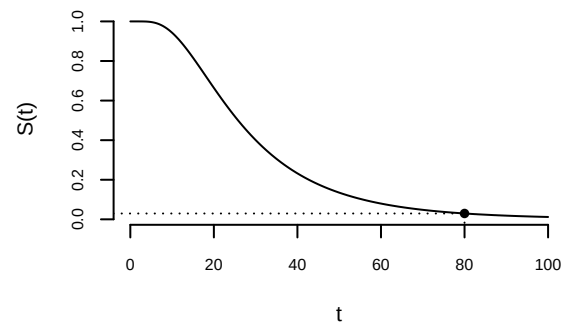
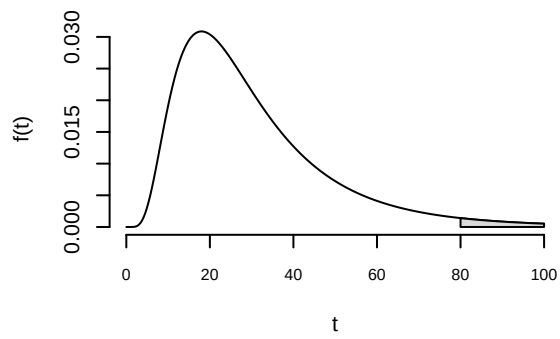
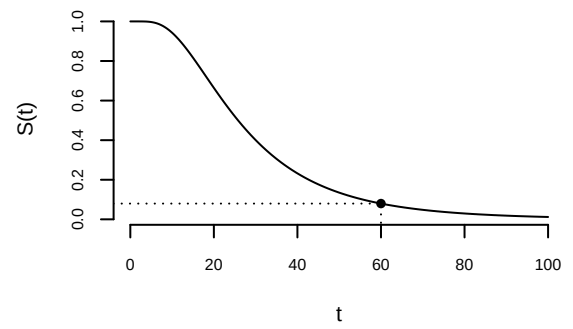
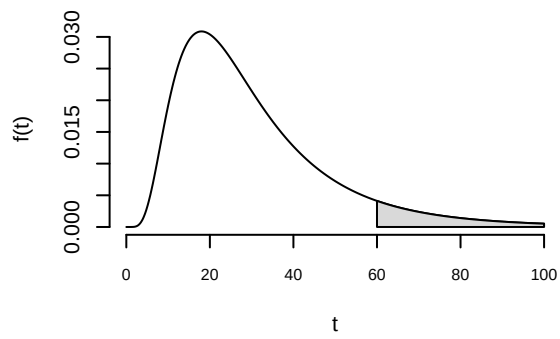
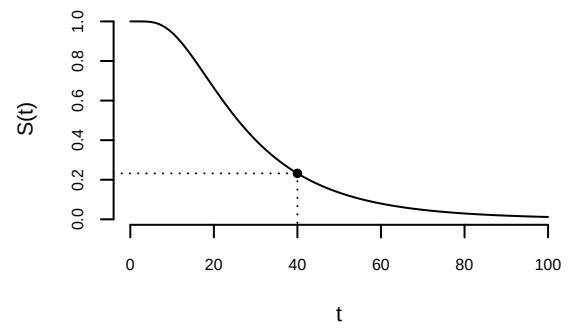
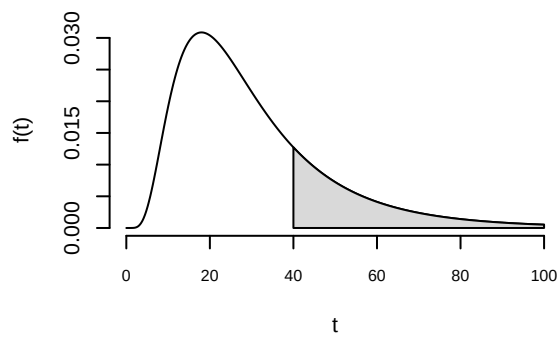
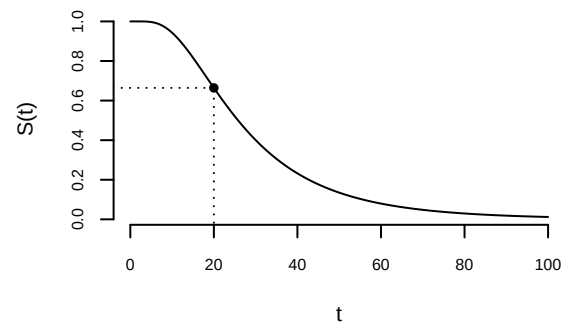
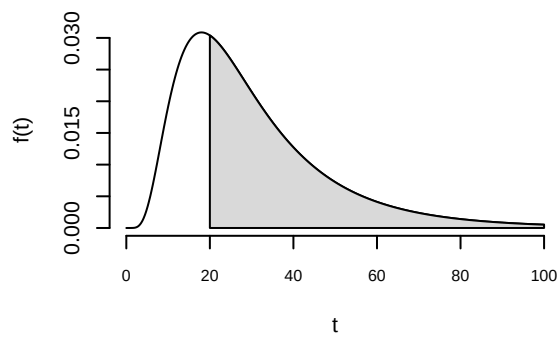


The Survival Function

The *survival function* is

$$S(t) = P(T \geq t).$$

It equals the area under $f(t)$ and between t and ∞ . The area under $S(t)$ equals $E(T)$ if $S(0) = 1$ and $S(\infty) = 0$.



Thus

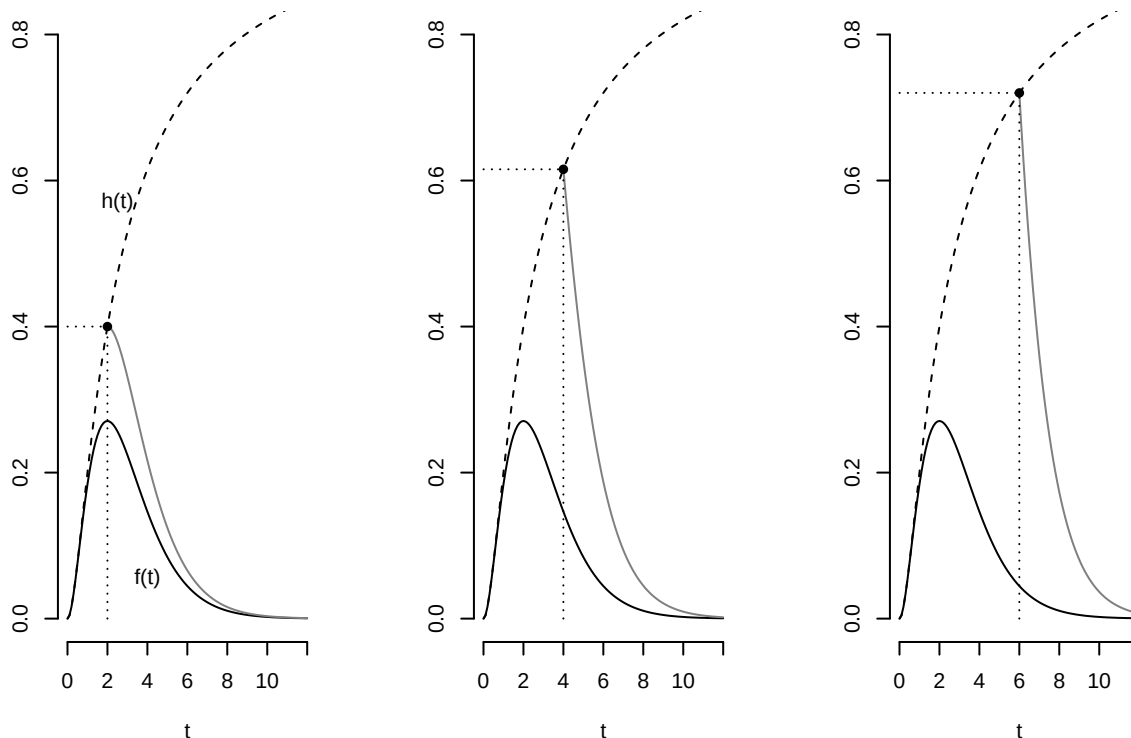
$$S(t) = \int_t^{\infty} f(z) dz.$$

The Hazard Function

The *hazard function* is

$$h(t) = \lim_{\delta t \rightarrow 0} \frac{P(t \leq T < t + \delta t | T \geq t)}{\delta t} = \frac{f(t)}{S(t)}.$$

If δ is relatively small then $h(t)$ is approximately *proportional* to the probability that $t \leq T < t + \delta t$ given survival up to t — i.e., $T \geq t$. So $h(t)$ is approximately proportional to the probability of the event happening at near time t if it has not yet happened.



Distributions and Hazard Functions

A wide variety of distributions can be used for parametric survival models such as AFT models. Below is a list of just some of those distributions. One of the more noticeable differences between them is the shape of their hazard functions.

1. *Log-normal*. The distribution of $\log(T_i)$ is normal. Single-peaked hazard function. Known as **lognormal** by **survreg** and **flexsurvreg**, and also **lnorm** by **flexsurvreg**.
2. *Log-logistic*. The distribution of $\log(T_i)$ is logistic. Single-peaked or decreasing hazard function. Known as **loglogistic** by **survreg** and **flexsurvreg** and **llogis** by **flexsurvreg**.
3. *Gamma*. Monotonic or flat hazard function. Known as **gamma** to **flexsurvreg**.
4. *Weibull*. Monotonic or flat hazard function. Known as **weibull** to both **survreg** and **flexsurvreg**.
5. *Exponential*. Flat hazard function (“memoryless”). Known as **exp** to **flexsurvreg** but also as a special case of **weibull** if **scale** = 1 with **survreg**.
6. *Gompertz*. Increasing hazard function. Known as **gompertz** to **flexsurvreg**.

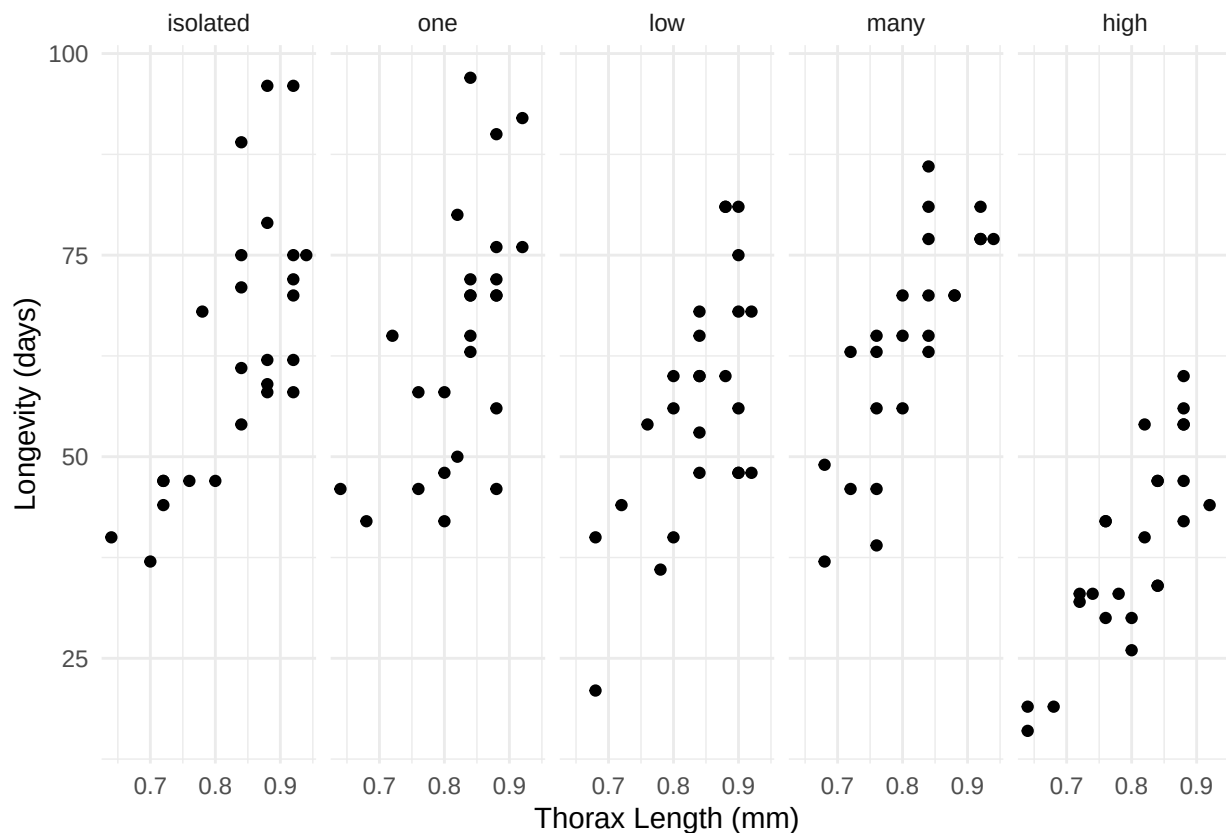
7. *Generalized gamma*. Monotonic, single-peaked, and “bathtub” hazard functions. The exponential, Weibull, gamma, and log-normal are special cases. Known as `gengamma` to `flexsurvreg`.
8. *Generalized F*. Single-peaked or decreasing. Known as `genf` to `flexsurvreg`.

Estimating and Plotting Hazard Functions

The `summary` function can be used to estimate the hazard function based on a `flexsurvreg` model object.

Example: Consider data from an experiment on the effects of sexual activity on the lifespan of the male fruitfly. Thorax length was used as a covariate.

```
library(faraway)
p <- ggplot(fruitfly, aes(x = thorax, y = longevity)) +
  geom_point() + facet_wrap(~ activity, ncol = 5) +
  labs(x = "Thorax Length (mm)", y = "Longevity (days)") +
  theme_minimal()
plot(p)
```



```
m <- flexsurvreg(Surv(longevity) ~ activity + thorax,
  data = fruitfly, dist = "gamma")

d <- data.frame(activity = levels(fruitfly$activity), thorax = 0.8)
d <- summary(m, newdata = d, t = seq(0, 100, length = 100),
  type = "hazard", tidy = TRUE)
head(d)
```

time	est	lcl	ucl	activity	thorax
------	-----	-----	-----	----------	--------

```

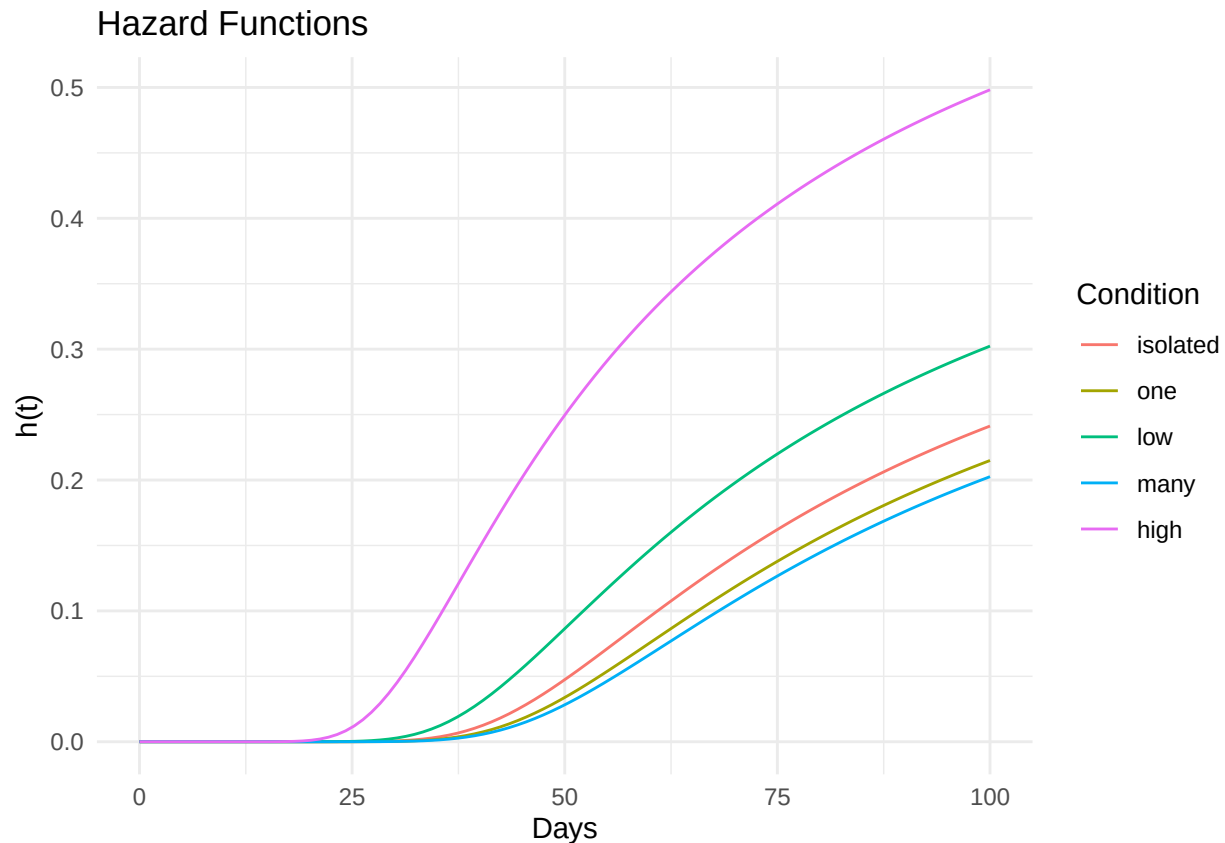
1 0.00 0.00e+00 0.00e+00 0.00e+00 isolated 0.8
2 1.01 1.27e-38 1.81e-49 6.83e-31 isolated 0.8
3 2.02 1.90e-30 6.57e-39 1.87e-24 isolated 0.8
4 3.03 9.30e-26 7.24e-33 9.47e-21 isolated 0.8
5 4.04 1.70e-22 1.04e-28 3.57e-18 isolated 0.8
6 5.05 5.14e-20 1.76e-25 3.24e-16 isolated 0.8

```

```

p <- ggplot(d, aes(x = time, y = est, color = activity)) +
  geom_line() + theme_minimal() +
  labs(x = "Days", y = "h(t)", color = "Condition", title = "Hazard Functions")
plot(p)

```

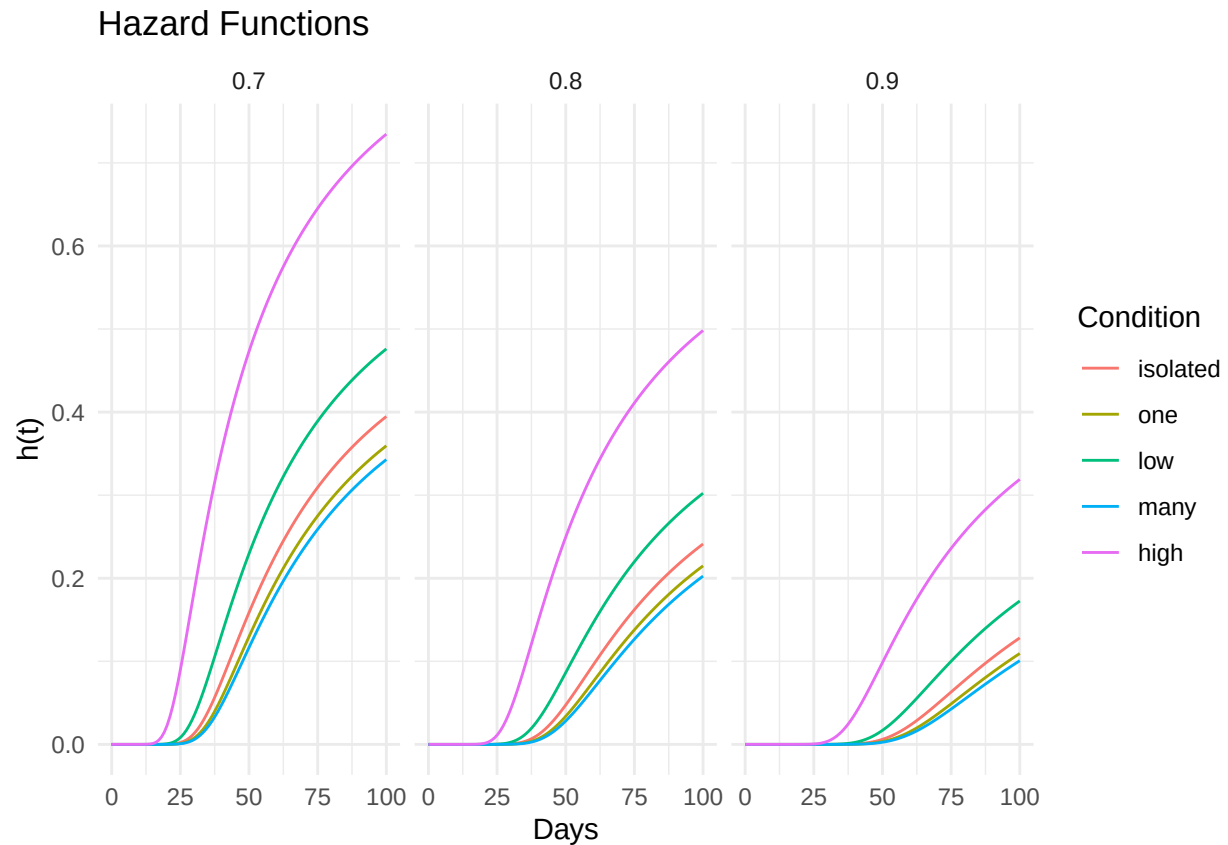


```

d <- expand.grid(activity = levels(fruitfly$activity), thorax = c(0.7,0.8,0.9))
d <- summary(m, newdata = d, t = seq(0, 100, length = 100),
  type = "hazard", tidy = TRUE)

p <- ggplot(d, aes(x = time, y = est, color = activity)) +
  geom_line() + theme_minimal() +
  labs(x = "Days", y = "h(t)", color = "Condition", title = "Hazard Functions") +
  facet_wrap(~ thorax)
plot(p)

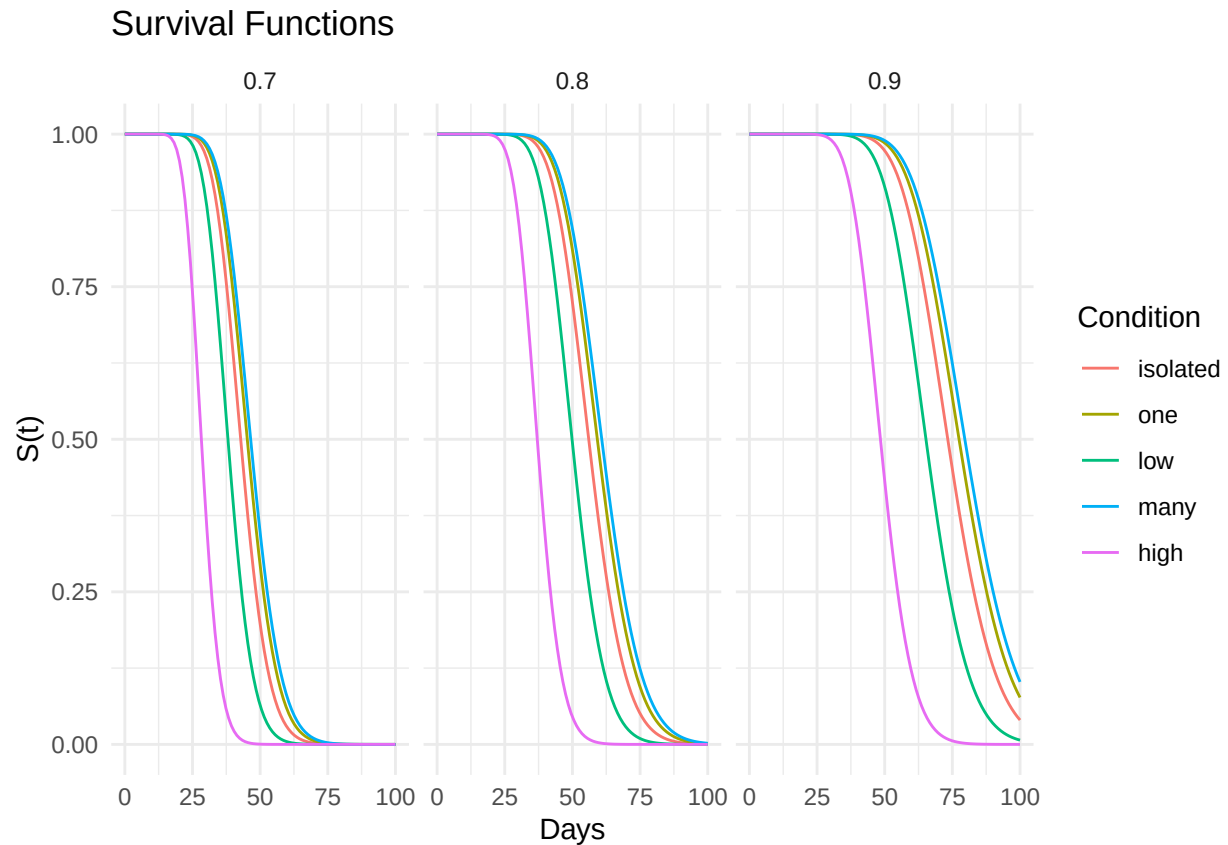
```



For comparison here are the *survival functions*.

```
d <- expand.grid(activity = levels(fruitfly$activity), thorax = c(0.7,0.8,0.9))
d <- summary(m, newdata = d, t = seq(0, 100, length = 100),
  type = "survival", tidy = TRUE)

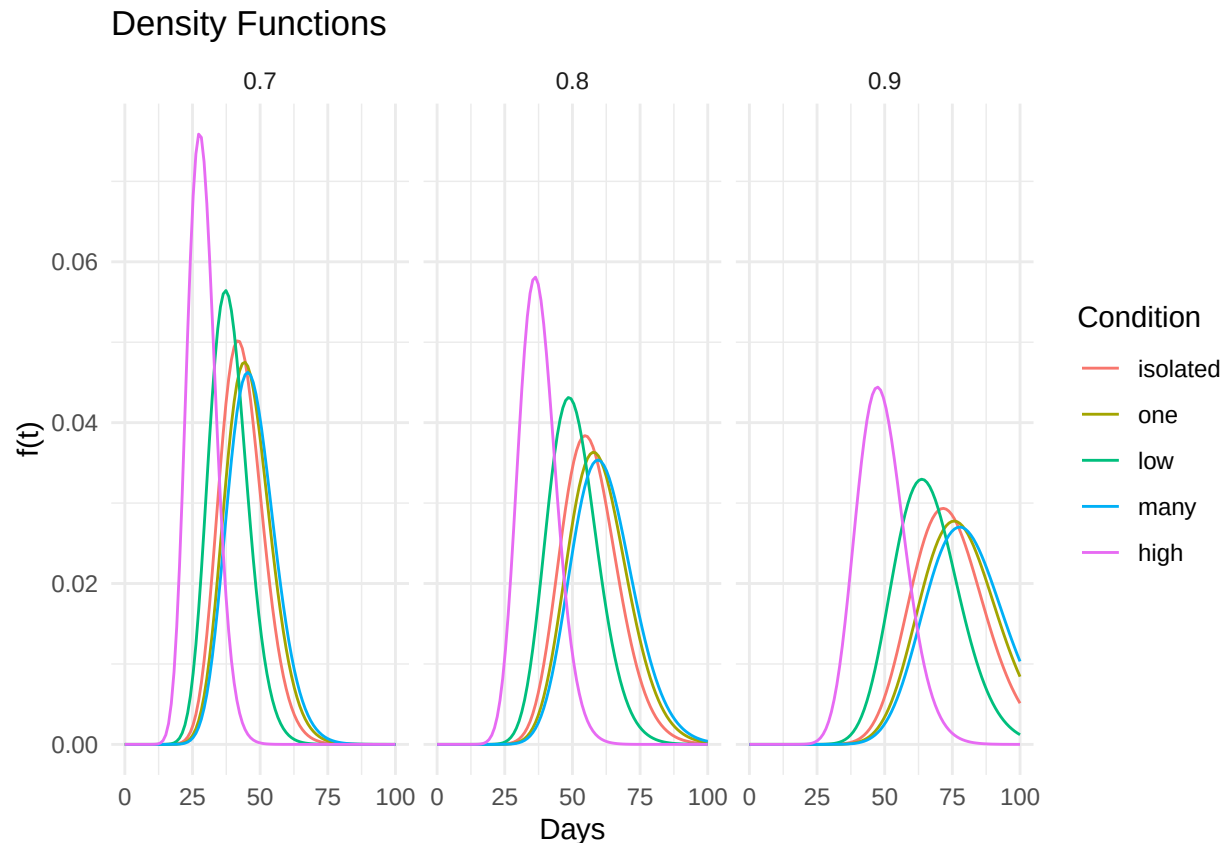
p <- ggplot(d, aes(x = time, y = est, color = activity)) +
  geom_line() + theme_minimal() +
  labs(x = "Days", y = "S(t)", color = "Condition", title = "Survival Functions") +
  facet_wrap(~ thorax)
plot(p)
```



And here are the *probability density functions*.

```
d <- expand.grid(activity = levels(fruitfly$activity), thorax = c(0.7,0.8,0.9))
d <- summary(m, newdata = d, t = seq(0, 100, length = 100),
  fn = function(t, ...) dgamma(t, ...), tidy = TRUE)

p <- ggplot(d, aes(x = time, y = est, color = activity)) +
  geom_line() + theme_minimal() +
  labs(x = "Days", y = "f(t)", color = "Condition", title = "Density Functions") +
  facet_wrap(~ thorax)
plot(p)
```



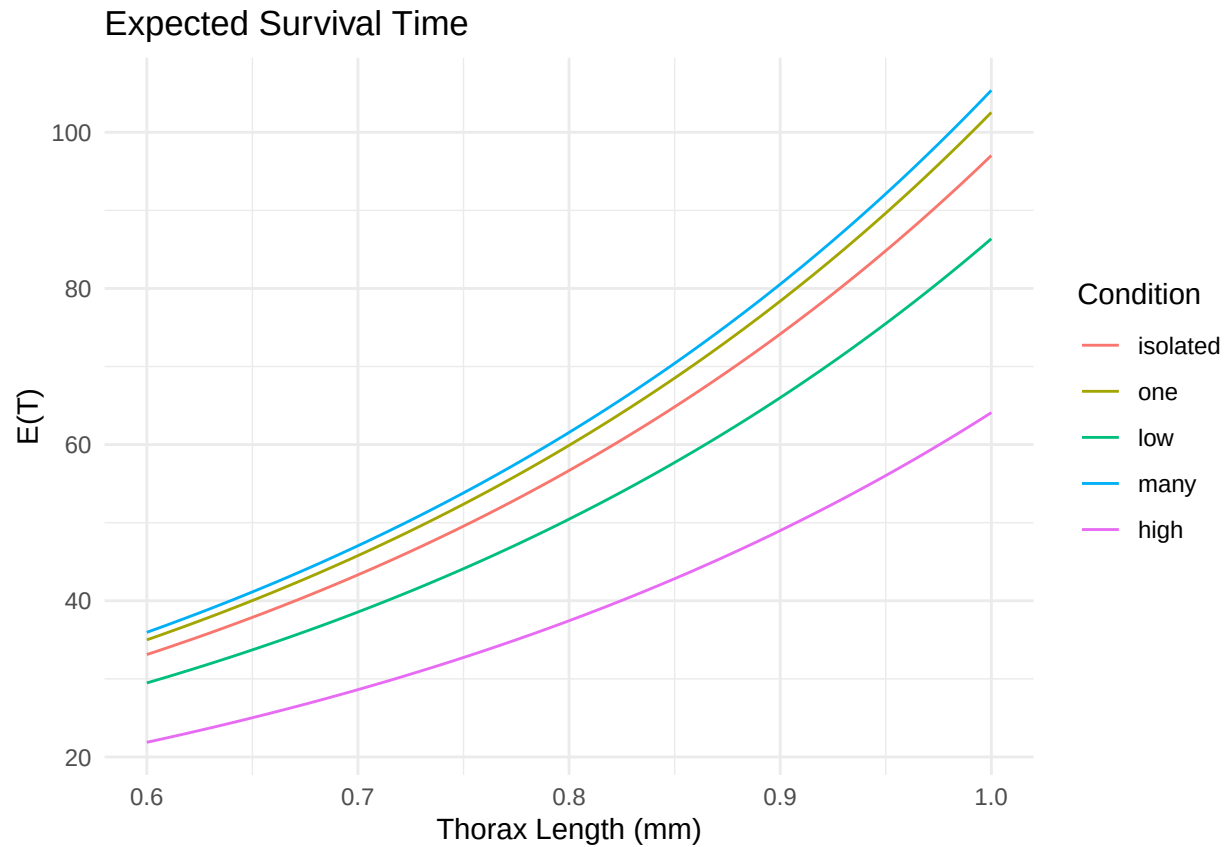
Note that we can adapt this to other distributions by adding a `d` to the beginning of the distribution name recognized by `flexsurvreg`. This includes log-normal (`dlnorm`), log-logistic (`dloglogis`), gamma (`dgamma`), Weibull (`dweibull`), exponential (`dexp`), Gompertz (`dgomperz`), generalized gamma (`dgengamma`), and generalized F (`dgenf`).

Finally we can also plot the *expected* survival time. This is analogous to using `predict` with `type = response` in a GLM.

```
d <- expand.grid(activity = levels(fruitfly$activity),
  thorax = seq(0.6, 1.0, length = 100))
d <- summary(m, newdata = d, type = "mean", tidy = TRUE)
head(d)
```

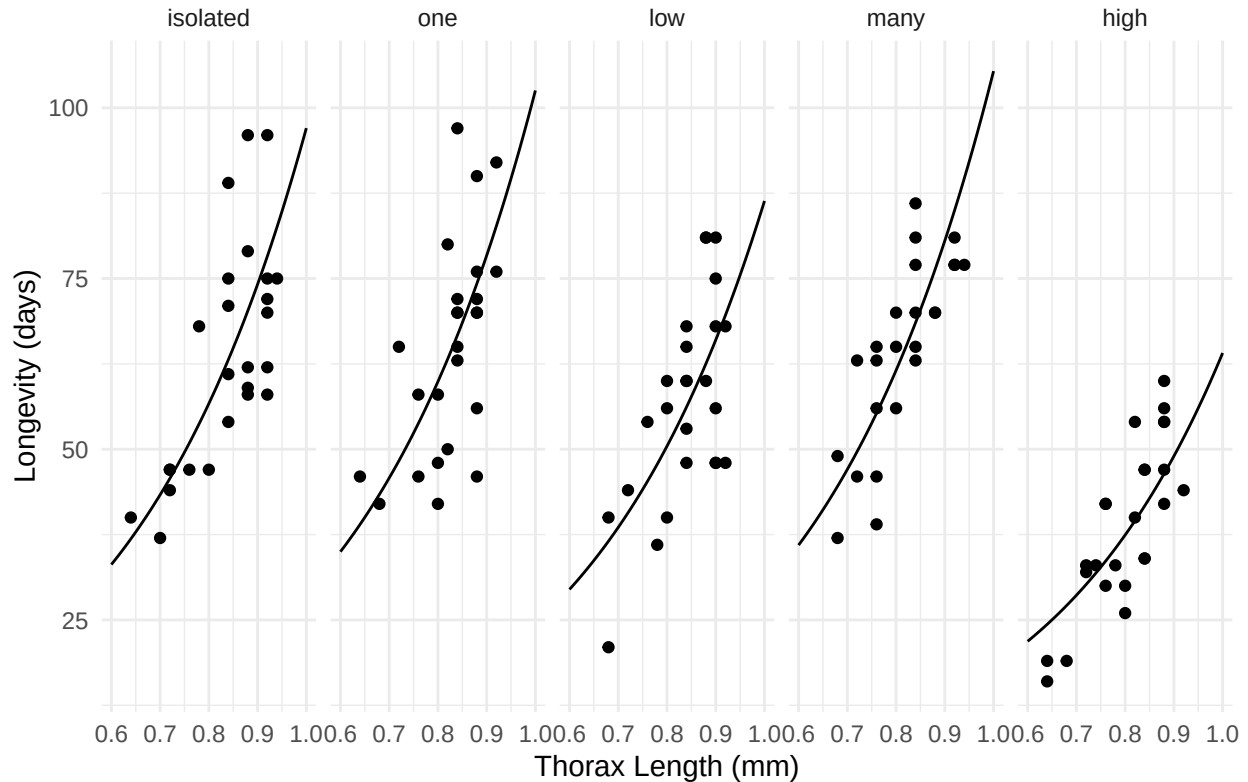
	est	lcl	ucl	activity	thorax
1	33.1	29.0	37.6	isolated	0.600
2	35.0	30.8	39.6	one	0.600
3	29.5	25.8	33.3	low	0.600
4	36.0	31.9	40.3	many	0.600
5	21.9	19.4	24.6	high	0.600
6	33.5	29.3	38.0	isolated	0.604

```
p <- ggplot(d, aes(x = thorax, y = est, color = activity)) +
  geom_line() + theme_minimal() +
  labs(x = "Thorax Length (mm)", y = "E(T)", color = "Condition",
    title = "Expected Survival Time")
plot(p)
```



```
p <- ggplot(fruitfly, aes(x = thorax, y = longevity)) +  
  geom_point() + facet_wrap(~ activity, ncol = 5) +  
  labs(x = "Thorax Length (mm)", y = "Longevity (days)",  
    title = "Observed and Expected Survival Time") +  
  theme_minimal() + geom_line(aes(y = est), data = d)  
plot(p)
```

Observed and Expected Survival Time



Example: Consider an AFT model for the leukemia data.

```
library(survival)
leukemia$status <- factor(leukemia$status, labels = c("alive", "dead"))

m <- flexsurvreg(Surv(time, status == "dead") ~ x,
  dist = "weibull", data = leukemia)

# create plot of hazard functions
d <- data.frame(x = c("Maintained", "Nonmaintained"))
d <- summary(m, newdata = d, t = seq(1, 200, length = 1000),
  type = "hazard", tidy = TRUE)

p <- ggplot(d, aes(x = time, y = est)) +
  geom_line(aes(linetype = x)) + theme_minimal() +
  labs(x = "Time", y = "h(t)", linetype = "Extended",
    title = "Hazard Functions") +
  theme(legend.position = "inside", legend.position.inside = c(0.7, 0.5))
p.h <- p

# create plot of survival functions
d <- data.frame(x = c("Maintained", "Nonmaintained"))
d <- summary(m, newdata = d, t = seq(1, 200, length = 1000),
  type = "survival", tidy = TRUE)

p <- ggplot(d, aes(x = time, y = est)) +
  geom_line(aes(linetype = x)) + theme_minimal() +
```

```

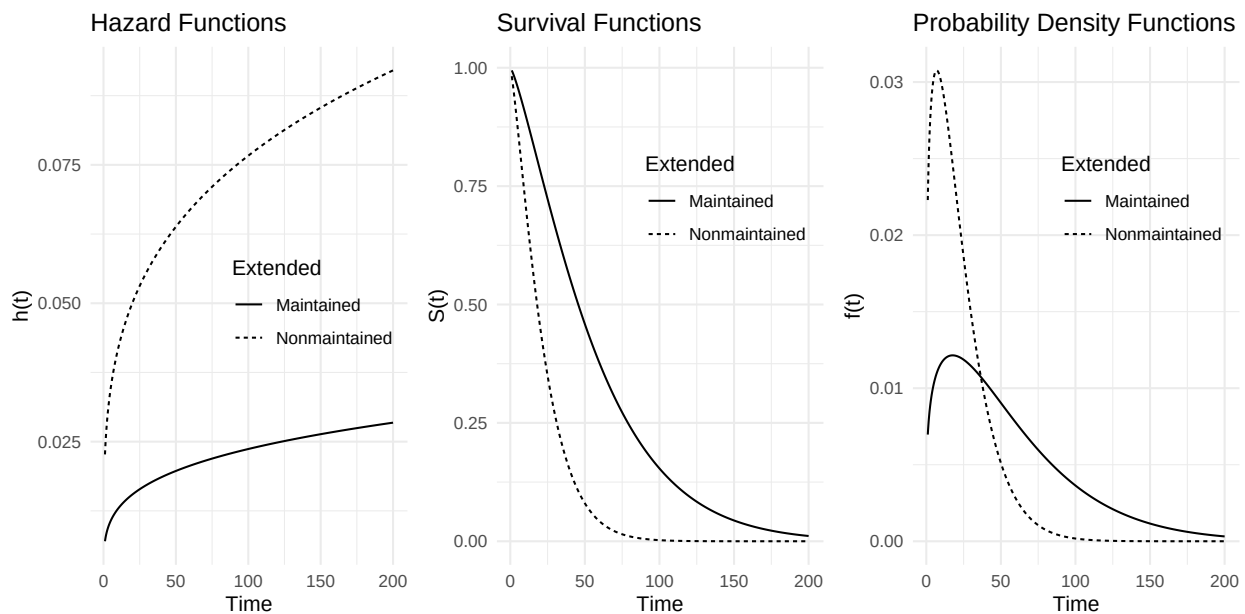
labs(x = "Time", y = "S(t)", linetype = "Extended",
     title = "Survival Functions") +
theme(legend.position = "inside", legend.position.inside = c(0.7, 0.7))
p.s <- p

# create plot of probability density functions
d <- data.frame(x = c("Maintained", "Nonmaintained"))
d <- summary(m, newdata = d, t = seq(1, 200, length = 1000),
             fn = function(t, ...) dweibull(t, ...), tidy = TRUE)

p <- ggplot(d, aes(x = time, y = est)) +
  geom_line(aes(linetype = x)) + theme_minimal() +
  labs(x = "Time", y = "f(t)", linetype = "Extended",
       title = "Probability Density Functions") +
  theme(legend.position = "inside", legend.position.inside = c(0.7, 0.7))
p.d <- p

# put the plots together into one plot
cowplot::plot_grid(p.h, p.s, p.d, ncol = 3)

```



We can also plot the raw data with the estimated expected survival times and confidence intervals for the estimated expected survival time.

```

d <- summary(m, newdata = data.frame(x = c("Maintained", "Nonmaintained")),
             type = "mean", tidy = TRUE)
d

```

```

  est  lcl  ucl      x
1 56.6 31.3 101.2 Maintained
2 22.3 14.4  37.4 Nonmaintained

```

```

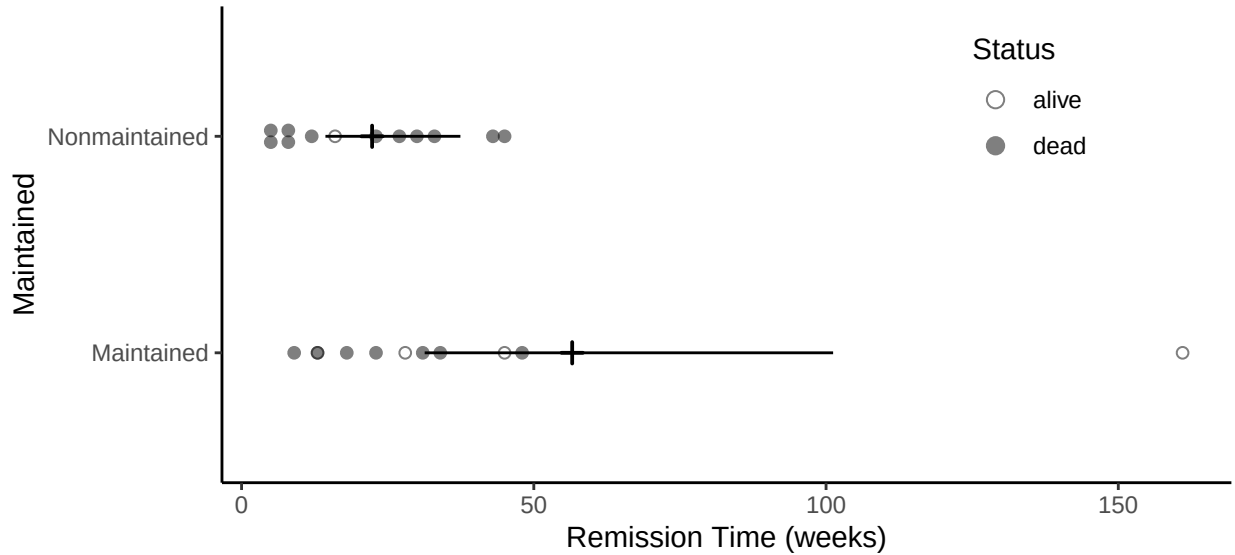
p <- ggplot(leukemia, aes(x = x, y = time)) +
  geom_dotplot(aes(fill = status), stackdir = "center", binaxis = "y",
               binwidth = 1, dotsize = 2, alpha = 0.5) + coord_flip() +

```

```

scale_fill_manual(name = "Status", values = c("white", "black")) +
geom_pointrange(aes(y = est, ymin = lcl, ymax = ucl),
  shape = 3, data = d) +
labs(x = "Maintained", y = "Remission Time (weeks)") +
theme_classic() +
theme(legend.position = "inside", legend.position.inside = c(0.8, 0.8))
plot(p)

```



A very useful feature of the **flexsurv** package is that a user can program their own distribution for use with the functions therein.

Proportional Hazards Models

Let $h_0(t)$ be the “baseline” hazard function (i.e., the hazard function when all $x_j = 0$). A *proportional hazards model* has the form

$$h_i(t) = h_0(t)e^{\beta_1 x_{i1}} e^{\beta_2 x_{i2}} \dots e^{\beta_k x_{ik}},$$

so that $h_i(t) \propto e^{\beta_1 x_{i1}} e^{\beta_2 x_{i2}} \dots e^{\beta_k x_{ik}}$. Thus increasing x_j by one changes the hazard function by a factor of e^{β_j} . This is the *hazard ratio*. For example, the hazard ratio for x_1 is

$$\frac{h_0(t)e^{\beta_1(x_1+1)} e^{\beta_2 x_2} \dots e^{\beta_k x_k}}{h_0(t)e^{\beta_1 x_1} e^{\beta_2 x_2} \dots e^{\beta_k x_k}} = e^{\beta_1},$$

since $e^{\beta_1(x_1+1)} = e^{\beta_1 x_1} e^{\beta_1}$.

Parametric Proportional Hazards Models

AFT models with a Weibull distribution (or exponential, which is a special case of the Weibull distribution) are also proportional hazards models. Consider the AFT model,

$$\log T_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_k x_{ik} + \sigma \epsilon_i,$$

and the proportional hazards model

$$h_i(t) = h_0(t) \exp(\beta_1^* x_{i1} + \beta_2^* x_{i2} + \dots + \beta_k^* x_{ik}),$$

where in both cases T_i has a *Weibull* distribution. It can be shown that the models are equivalent with

$$\beta_j^* = -\beta_j / \sigma.$$

The hazard ratios are $e^{\beta_j^*}$.

An AFT model with a Weibull distribution is the *only* AFT model that is also a proportional hazards model. Other proportional hazards models exist, but none of them are AFT models.

Example: We can estimate a Weibull proportional hazards model for the leukemia data using `survreg` as follows.

```
m <- survreg(Surv(time, status == "dead") ~ x, dist = "weibull", data = leukemia)
summary(m)
```

Call:

```
survreg(formula = Surv(time, status == "dead") ~ x, data = leukemia,
        dist = "weibull")
```

	Value	Std. Error	z	p
(Intercept)	4.109	0.300	13.70	<2e-16
xNonmaintained	-0.929	0.383	-2.43	0.015
Log(scale)	-0.235	0.178	-1.32	0.188

Scale= 0.791

Weibull distribution

Loglik(model)= -80.5 Loglik(intercept only)= -83.2

Chisq= 5.31 on 1 degrees of freedom, p= 0.021

Number of Newton-Raphson Iterations: 5

n= 23

The estimated hazard ratio is $e^{\hat{\beta}_1^*}$ where $\hat{\beta}_1^* \approx 0.929/0.791 \approx 1.175$ so $e^{\hat{\beta}_1^*} \approx 3.238$. Thus

$$\frac{h_n(t)}{h_y(t)} = e^{\beta_1^*} \Leftrightarrow h_n(t) = e^{\beta_1^*} h_y(t),$$

where we estimate the hazard ratio $e^{\beta_1^*}$ to be 3.238. This conversion can be done using the `ConvertWeibull` function from the `SurvRegCensCov` package.

```
library(SurvRegCensCov)
ConvertWeibull(m)
```

\$vars

	Estimate	SE
lambda	0.00554	0.0057
gamma	1.26430	0.2253
xNonmaintained	1.17496	0.5230

\$HR

	HR	LB	UB
xNonmaintained	3.24	1.16	9.03

\$ETR

	ETR	LB	UB
xNonmaintained	0.395	0.187	0.836

Another approach is to use `dist = "weibullPH"` with `flexsurvreg` which uses a different parameterization of the Weibull distribution so that applying the exponential function to the parameters gives hazard ratios.

```
m <- flexsurvreg(Surv(time, status == "dead") ~ x, dist = "weibullPH", data = leukemia)
print(m)
```

```
Call:
flexsurvreg(formula = Surv(time, status == "dead") ~ x, data = leukemia,
  dist = "weibullPH")
```

Estimates:

	data	mean	est	L95%	U95%	se	exp(est)	L95%	U95%
shape	NA		1.264295	0.891546	1.792889	0.225328	NA	NA	NA
scale	NA		0.005544	0.000739	0.041565	0.005698	NA	NA	NA
xNonmaintained	0.521739		1.174962	0.149832	2.200092	0.523035	3.238021	1.161640	9.025845

N = 23, Events: 18, Censored: 5

Total time at risk: 678

Log-likelihood = -80.5, df = 3

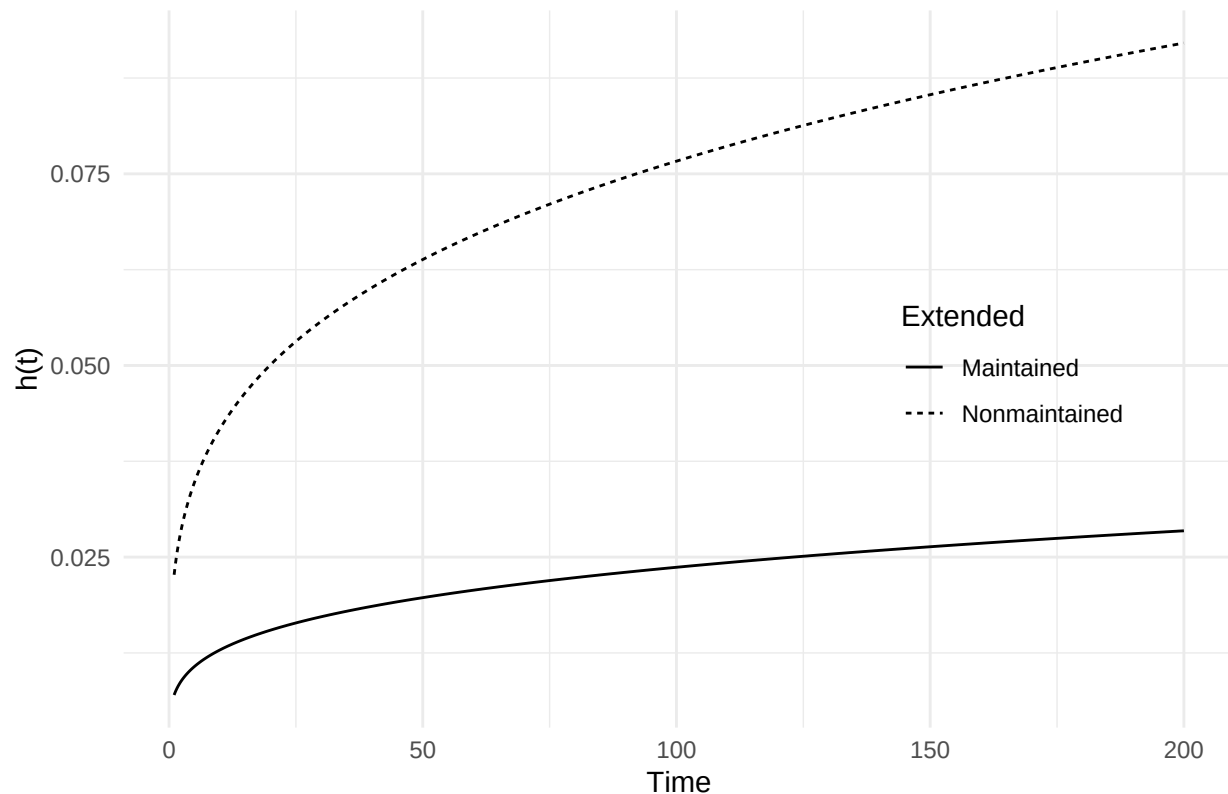
AIC = 167

The proportionality can be seen when plotting the hazard functions.

```
d <- data.frame(x = c("Maintained", "Nonmaintained"))
d <- summary(m, newdata = d, t = seq(1, 200, length = 1000),
  type = "hazard", tidy = TRUE)

p <- ggplot(d, aes(x = time, y = est)) +
  geom_line(aes(linetype = x)) + theme_minimal() +
  labs(x = "Time", y = "h(t)", linetype = "Extended", title = "Hazard Functions") +
  theme(legend.position = "inside", legend.position.inside = c(0.8, 0.5))
plot(p)
```

Hazard Functions



Example: Consider a Weibull proportional hazards model for the motors data.

```
m <- flexsurvreg(Surv(time, cens) ~ temp, data = MASS::motors, dist = "weibullPH")
print(m)
```

Call:

```
flexsurvreg(formula = Surv(time, cens) ~ temp, data = MASS::motors,
  dist = "weibullPH")
```

Estimates:

	data	mean	est	L95%	U95%	se	exp(est)	L95%	U95%
shape	NA		2.99e+00	1.96e+00	4.56e+00	6.42e-01	NA	NA	NA
scale	NA		6.34e-22	1.46e-30	2.76e-13	6.43e-21	NA	NA	NA
temp	1.82e+02		1.36e-01	7.92e-02	1.92e-01	2.87e-02	1.15e+00	1.08e+00	1.21e+00

N = 40, Events: 17, Censored: 23

Total time at risk: 140654

Log-likelihood = -147, df = 3

AIC = 301

Here we have that

$$h_{x+1}(t) = e^{\beta_1^*} h_x(t),$$

where $h_x(t)$ and $h_{x+1}(t)$ represent the hazard functions at temperatures of x and $x + 1$, respectively. The estimated hazard ratio is $e^{\beta_1^*} = 1.15$.

```
d <- summary(m, newdata = data.frame(temp = seq(110, 150, by = 10)),
  t = seq(0, 8000, length = 1000), type = "hazard", tidy = TRUE, ci = FALSE)

p <- ggplot(d, aes(x = time, y = est, color = factor(temp))) +
  geom_line() + theme_minimal() +
  theme(legend.position = "inside", legend.position.inside = c(0.2, 0.6)) +
  labs(x = "Hours", y = "h(t)", color = "Temperature", title = "Hazard Functions")
plot(p)
```

Hazard Functions

